TEXORA · LECTURE SHEET

# Material Requirement Planning (MRP) and ERP

*Planning dependent demand — how MRP explodes a master schedule through the bill of materials into timed net requirements and planned orders, and how ERP widens that logic to the whole enterprise.*

TE-4107 · PRODUCTION PLANNING AND CONTROL · TOPIC 6 · LEVEL 4 TERM 1

## 1. Why MRP exists: dependent demand

Some demands are **independent** — finished shirts ordered by buyers — and are forecast or booked directly. Most factory demand is **dependent**: the fabric for those shirts, the yarn for that fabric, the buttons, thread and cartons are needed *because* the shirts are needed, in quantities fixed by the product structure. Planning dependent items with independent-demand methods (reorder points, guesswork) produces the classic textile paradox: stores full of materials and production stopped for want of one of them. **Material Requirement Planning (MRP)** replaces guessing with arithmetic: from the master production schedule (MPS), the bill of materials (BOM) and inventory records, it computes exactly what must be made or bought, how much, and by when.

## 2. Objectives and functions of MRP

- Ensure every component is available when its parent operation starts — no stoppages for materials.
- Keep inventory at the minimum consistent with that availability — dependent items are ordered for need, not for comfort.
- Schedule by priority: when plans change, MRP re-dates requirements so that genuinely urgent orders are expedited and premature ones are pushed back, instead of everything being "urgent".
- Provide a feasible material plan that capacity planning (Topic 1) can test — MRP says what materials the schedule needs; capacity says whether the schedule itself is achievable.

## 3. The MRP system: inputs, processing, outputs

| INPUTS | WHAT IT CONTAINS | TEXTILE FORM |
|---|---|---|

| | | |
|---|---|---|
| Master production schedule (MPS) | What finished products, how many, by when | Confirmed order book + forecast, by style/order and delivery week |
| Bill of materials (BOM) | Product structure: components per parent, level by level | Consumption norms: fabric m/pc, yarn kg/m, trims per garment, process wastage % |
| Inventory records | Stock on hand, scheduled receipts (open orders), lead times, lot rules | Store ledgers / ERP stock of fabric, yarn, dyes, trims; purchase orders in transit |

**Processing** works level by level through the BOM — **explosion**. For each item in each period: **Gross requirement** (from parent planned orders) − **on hand** − **scheduled receipts** = **net requirement**; net requirements are lot-sized (exact, fixed lot, or period cover) into **planned orders**, then **offset backwards by lead time** to give planned order *releases*. A parent's releases generate the gross requirements of its children — the explosion cascades to raw materials.

**Outputs:** planned order releases for production and purchase (what to start/buy, when), reschedule and cancellation notices for open orders, and exception reports (late, overdue, impossible dates) — the planner's daily action list.

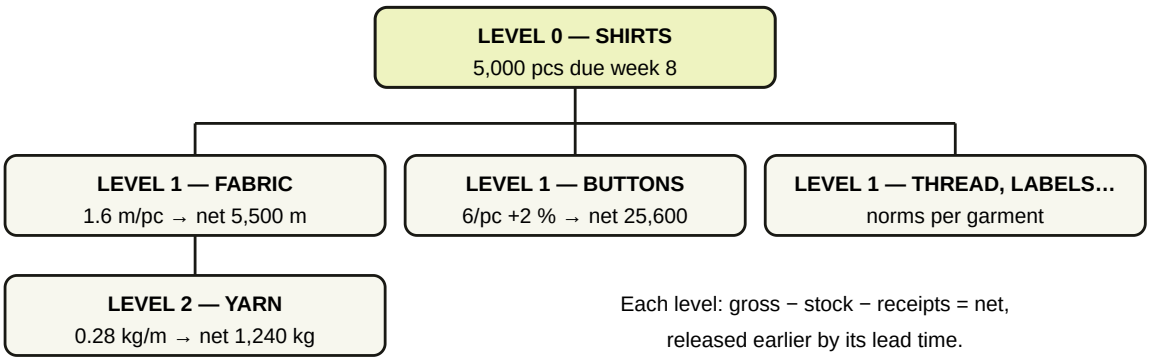



Fig. 1 — BOM explosion for the worked example: the shirt order drives fabric and trims at level 1 and yarn at level 2.

**WORKED EXAMPLE — MRP EXPLOSION FOR A SHIRT ORDER**

Order: **5,000 shirts**, delivery week 8. Sewing starts week 6 (lead time 2 weeks from material availability).

**Fabric** (BOM 1.6 m/shirt): gross = 5,000 × 1.6 = 8,000 m. On hand 1,500 m; scheduled receipt 1,000 m (open weaving order). Net = 8,000 − 1,500 − 1,000 = **5,500 m**, to be ready by week 6; weaving lead time 2 weeks → release weaving order in **week 4**.

**Yarn for that fabric** (0.28 kg/m): gross = 5,500 × 0.28 = 1,540 kg. On hand 300 kg. Net = **1,240 kg**, needed by week 4; purchase/spinning lead time 3 weeks → release in **week 1**.

**Buttons** (6/shirt + 2 % process allowance): gross = 5,000 × 6 × 1.02 = 30,600. On hand 5,000. Net = **25,600 buttons**, needed by week 6; purchase lead time 4 weeks → order in **week 2**.

Note the direction of time: the customer date is in week 8, yet the first action — releasing yarn —

> falls in week 1. Lead-time offsetting is what turns a delivery promise into a calendar of start dates; without it, every order is discovered to be late exactly when it is too late.

## 4. ERP: from MRP to the whole enterprise

**Enterprise Resource Planning (ERP)** generalises MRP's single database logic to all functions: sales, production, materials, quality, maintenance, HR and finance work on one integrated database, so a confirmed order simultaneously creates production demand, material reservations, capacity load and a receivable schedule. **Working areas** in a textile ERP typically include order management and costing, production planning and floor reporting, inventory and purchase, dye-house recipe and batch management, quality records, payroll and accounts. **Functions and benefits:** one version of the truth (no conflicting spreadsheets), transaction discipline (stock moves only against documents — MRP is only as good as inventory accuracy), traceability from bale to carton, and management reporting from live data. The cautions are equally standard: ERP enforces the process it is configured with — a bad process becomes a faithfully enforced bad process — and data accuracy (BOM norms, stock records, lead times) must be maintained as a daily discipline, not an implementation-week event. MRP inside an ERP with 80 %-accurate stock records produces precisely calculated wrong answers.

## Glossary

**Independent demand** — demand for finished goods, forecast/booked directly. **Dependent demand** — component demand derived from parent demand via the BOM. **MPS** — master production schedule of finished products. **BOM** — bill of materials; the product structure with consumption norms. **Explosion** — level-by-level computation of component requirements from parent planned orders. **Net requirement** — gross requirement minus on-hand and scheduled receipts. **Lead-time offsetting** — releasing an order one lead time before its requirement date. **ERP** — enterprise-wide integrated planning and transaction system built around one database.

## Key points

• MRP exists because component demand is dependent: it is calculated from the MPS through the BOM, never forecast item by item.
• Core arithmetic: net = gross − on hand − scheduled receipts; release = need date − lead time.
• Worked explosion: 5,000 shirts → 5,500 m net fabric (release week 4) → 1,240 kg net yarn (release week 1) and 25,600 net buttons (order week 2).
• ERP extends the same single-database discipline to the whole enterprise — and faithfully amplifies bad data or bad processes.

## Practice

1. An order for 8,000 polo shirts needs fabric at 1.9 m/pc. On hand 2,200 m, scheduled receipt 3,000 m. Compute net fabric. If fabric needs yarn at 0.30 kg/m with 500 kg on hand, compute net yarn.
2. Why does MRP use net requirements rather than simply ordering the gross quantities? State two business consequences of ignoring stock and scheduled receipts.
3. A buyer's order is confirmed 5 weeks before delivery. Sewing takes 1 week, fabric weaving 2 weeks, yarn purchase 3 weeks. Show, week by week counting back, whether the order is feasible from purchased yarn, and what the planner's options are if not.
4. "Our ERP is installed, so our planning is now correct." Give two reasons this may be false, with textile examples.

### Answer key

1. Gross fabric = 8,000 × 1.9 = 15,200 m. Net = 15,200 − 2,200 − 3,000 = 10,000 m. Yarn gross = 10,000 × 0.30 = 3,000 kg; net = 3,000 − 500 = 2,500 kg.
2. Because part of the gross need is already covered — by stock on the shelf and by orders already placed and arriving. Ordering gross would duplicate both: working capital locked in surplus fabric and yarn, store congestion, shade/count obsolescence when styles change — while, paradoxically, a planner who "knows" stock is roughly sufficient may under-order a different item and stop production. Netting makes each purchase exactly the uncovered need, in the right week.
3. Counting back from delivery (end of week 5): sewing occupies week 5 (needs fabric at start of week 5); weaving 2 weeks occupies weeks 3–4 (needs yarn at start of week 3); yarn purchase 3 weeks must be released at start of week 0 — before the order was confirmed. Infeasible from purchased yarn. Options: use yarn/fabric already in stock or in the pipeline (netting may save the order); buy yarn from a local source at shorter lead time or air-freight the first lot; split the order — sew the first part from stock fabric, the balance as materials arrive, with buyer's agreement; negotiate a later or split delivery. The planner's job is to present these options with dates and costs, not to discover the problem in week 4.
4. (i) Data accuracy: if BOM consumption norms are outdated (say fabric norm 1.6 m/pc while cutting actually uses 1.75) or stock records are wrong, ERP computes precise plans from false inputs — garbage in, gospel out. (ii) Process fit: if the configured workflow does not match how the dye house actually batches or how orders are actually confirmed, staff keep parallel spreadsheets and the "single database" quietly stops being single or current.